

AI-Based IGRT with 2D Image-Guided Dose Tracking of Daily Cone-Beam CTs for Assessing Prostate Position, Rectum, and Bladder Filling in Definitive Radiotherapy of Prostate Carcinoma

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Abstract

A method for the automatic detection of the bladder and rectum in daily Cone-Beam CTs (kV-CBCTs) for image-guided radiotherapy (IGRT) is presented. Medial sagittal slices – centered through the symphysis and one each to the left and right – were segmented using artificial intelligence (AI) to evaluate organ position and the filling states of the rectum and bladder during definitive radiotherapy of prostate carcinoma. The model was trained and validated with data from 131 patients (5 CBCTs per patient). The results demonstrate high segmentation accuracy with Dice coefficients of approximately 0.98 for the bladder and prostate PTV, and 0.95 for the rectum.

1 Introduction

In the definitive irradiation of low- and intermediate-risk prostate carcinomas, curative doses of 68 Gy are applied in moderate hypofractionation at our institution [Kuhnt and Nägler, 2024]. Intrafractional variability, such as filling states or organ motion, can lead to risk organs like the bladder and rectum receiving undesirably high doses, potentially increasing acute or chronic toxicity [Kupelian et al., 2007, Marks et al., 2010]. Image-guided radiotherapy (IGRT) with Cone-Beam CTs (CBCTs) verifies organ positions before each fraction. However, varying filling states – such as a poorly filled bladder or an air-filled rectum – complicate adherence to dose constraints like $V_{65} < 50\%$ for the

bladder [Wang et al., 2022, Marks et al., 2010]. Online adaptive radiotherapy enables daily adaptation to these changes but requires significant personnel effort (15–35 minutes per fraction) [Byrne et al., 2022], which is hardly feasible in high-throughput clinics. Often, only an offline review is performed, identifying deviations only after the fact. This study develops an AI-based method for partially automating online adaptive therapy, inspired by advances in medical image segmentation [Litjens et al., 2017], to more efficiently monitor prostate position and organ filling and improve treatment quality.

2 Methods

Between July 1, 2019, and January 31, 2024, 131 patients with low- to intermediate-risk prostate carcinoma who received definitive, moderately hypofractionated radiotherapy (25 fractions, 58–68 Gy) were retrospectively selected [Kuhnt and Nägler, 2024]. Inclusion criteria included cT1a–cT2b, $\text{PSA} \leq 20 \text{ ng/ml}$, and Gleason Score ≤ 7 ; patients with high-risk profiles or distant metastases were excluded. Written consent for anonymized data use was obtained.

For each patient, a planning CT (PL-CT) and 5 kV-CBCTs were anonymously extracted from the RayStation system (versions 10B–12A) [Zwart et al., 2022]. Structures (prostate PTV, bladder, rectum, etc.) were manually contoured by a medical physicist and reviewed by a specialist physician. A Python program (*sqlite_dicom2.py*) converted DICOM data into volumes, extracted three medial sagittal 2D slices per CBCT – a central slice through the symphysis and one each to the left and right – and generated binary masks of the contours using *scipy.ndimage.ConvexHull*. The outputs (patient number, DICOM series, slice images, mask, slice number) were stored in an SQLite database. A second program (*sqlite_viewer.py*) facilitated visual inspection and data cleaning.

The images were preprocessed (removal of unusually sized images, normalization of pixel data). In total, approximately 2100 images per organ were generated ($3 \text{ slices} \times 5 \text{ CBCTs} \times 131 \text{ patients}$), of which about 1700 were used for training and 400 for validation. Patients whose images were used for training were excluded from the validation dataset. To enhance quality, the 1700 training images were augmented – random rotation, scaling, translation, and noise – resulting in approximately 9000 images, a proven method for improving model robustness in image segmentation [Ronneberger et al., 2015, Pereira et al., 2016, Shorten and Khoshgoftaar, 2019].

A Convolutional Neural Network (CNN) based on U-Net [Ronneberger et al., 2015] was trained using PyTorch, accelerated by the ROCm stack on an AMD RX 7900 XTX GPU with 24 GB VRAM. The model consists of an encoder-decoder architecture with convolutional blocks (each with two 3×3 convolutions with ReLU activation) and transposed convolutions for upsampling layers. It was trained with a learning rate of 10^{-4} and the Adam optimizer over 25 epochs, using binary cross-entropy with logits (*BCEWithLogitsLoss*) as the loss function. Only CBCT data were used for training to ensure consistency [Litjens et al., 2017]. Training data were organized in a custom *ImageDataset* as PyTorch

tensors (stacked from Pandas Series) and loaded with a *DataLoader* (batch size 32, 4 workers, *pin_memory=True*). During training, images and masks were moved to the GPU in batches, with the loss backpropagated and optimized after each batch. Training time was approximately 1 hour per organ.

Bladder volume was roughly estimated by measuring the height and width of the bladder mask in the central sagittal plane and calculating the volume of a prolate spheroid. For the rectum, the volume of a cylinder was used for estimation.

3 Results

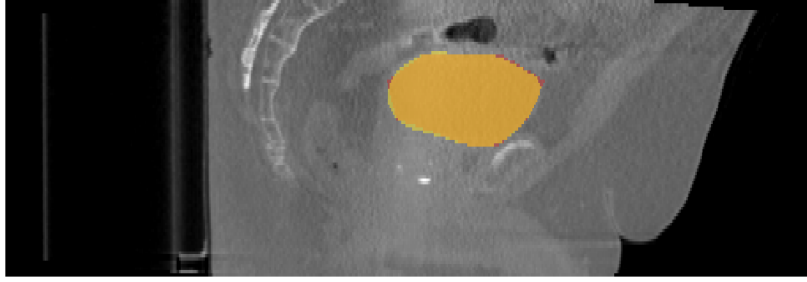
The U-Net model was tested on approximately 400 validation images per organ. The segmentation accuracy yielded the following Dice coefficients, averaged over all validation images:

Organ	Dice Coefficient
Bladder	0.9815 ± 0.0126
Prostate PTV	0.9789 ± 0.0092
Rectum	0.9552 ± 0.1021

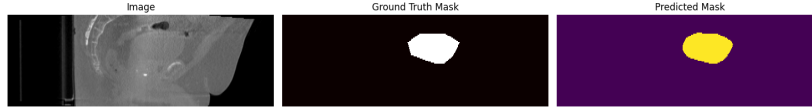
Table 1: Dice coefficients with standard deviations for various organs.

The volumes of the bladder and rectum were estimated. These values can be compared with the volumes from the PL-CT to assess filling states.

Manually created contours, reviewed by a specialist physician, served as the "Ground Truth." Model predictions were visually compared with the Ground Truth data. Examples are shown in Figures 1, 2, and 3.



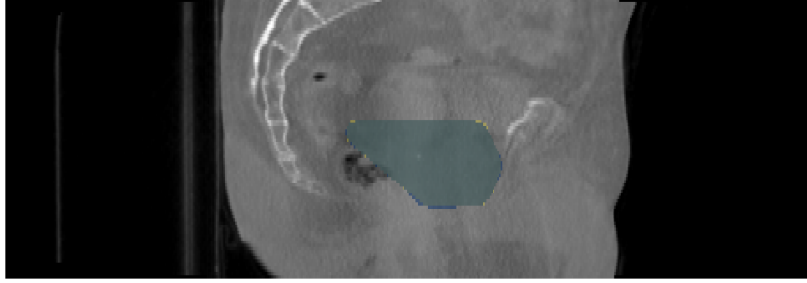
(a) Example segmentation of the bladder. Orange: Ground Truth, Yellow: Prediction.



(b) Example segmentation of the bladder. Left: Original image, Center: Ground Truth, Right: Model prediction.

Figure 1: Overview of segmentation results for the bladder.

In Figure 1b, the segmentation visibly covers the bladder better than the Ground Truth, which appears jagged due to segmentation artifacts. The model prediction is smoother and fully encompasses the bladder. The bladder filling in this example was estimated at 150 ml.



(a) Example segmentation of the prostate PTV. Yellow: Ground Truth, Blue: Prediction.

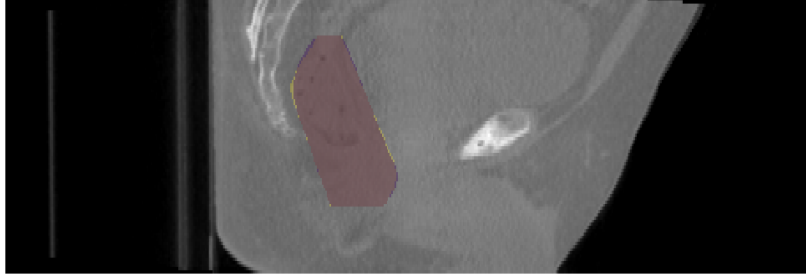


(b) Example segmentation of the prostate PTV. Left: Original image, Center: Ground Truth, Right: Model prediction.

Figure 2: Overview of segmentation results for the prostate PTV.

The model prediction in Figure 2b covers the prostate PTV very well. This

is of optional interest, as the PTV lacks clearly defined anatomical boundaries like the bladder. Nevertheless, the actual PTV can be overlaid on the CBCT to estimate the dose delivered to risk organs.



(a) Example segmentation of the rectum. Yellow: Ground Truth, Brown: Prediction.



(b) Example segmentation of the rectum. Left: Original image, Center: Ground Truth, Right: Model prediction.

Figure 3: Overview of segmentation results for the rectum.

The results for the rectum in Figure 3b are also satisfactory, though less precise than for the bladder and prostate PTV. This is reflected in the lower Dice coefficient and higher standard deviation. The rectum filling in this example was estimated at 300 ml.

4 Discussion

The method enables precise segmentation of the bladder and rectum. The Dice coefficient of approximately 0.98 for the bladder and prostate PTV provides reliable results for daily use. The value of approximately 0.95 for the rectum is also acceptable, particularly in conjunction with visualization. The higher standard deviation of approximately 0.1 for the rectum reflects its greater variability [Litjens et al., 2017]. The trained models can be executed on a standard computer in a few seconds (near real-time), enabling integration into clinical workflows, such as a plugin for the RayStation system.

Extending the model to the full kV-CBCT volume would allow 3D segmentation of risk organs, enhancing the method’s accuracy and robustness. Direct measurement of organ volumes would be particularly useful. However, this would likely require a larger dataset and more computational power.

Limitations of the method include dependence on consistent CBCT parameters and the inaccuracy due to information loss from dimensionality reduction

to 2D.

Future work could employ transfer learning or advanced augmentation to improve robustness against variable devices.

5 Conclusion

The AI-based method effectively automates the verification of prostate position and organ filling, offering potential to enhance treatment quality and efficiency in IGRT. The high Dice coefficients support the possibility of use without fiducials (e.g., gold markers) in the prostate, as is currently common. It lays the foundation for broader clinical applications following further validation.

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7 References

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